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**Economics 104**  
**Data Science for Economists**  
**Winter, 2024**

**Midterm Exam**  
**February 22, 2024**

|                   |  |
|-------------------|--|
| <b>First Name</b> |  |
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| <b>TA/Section</b> |  |

There are 20 multiple choice questions and all are worth the same number of points.  
For full credit on a problem, you need to show all your work and the formula(s) used.

1. The Boruta output below is based on the `HousePrices` dataset which we use to estimate a model for the price of a home according to:

$$price = \beta_0 + \beta_1 lotsize + \beta_2 bedrooms + \beta_3 bathrooms + \beta_4 stories + \beta_5 gasheat + \beta_6 garage + \beta_7 aircon + e.$$

Based on the Boruta output, which variables rank as the top 3 most important ones?

|           | meanImp | medianImp | minImp | maxImp | normHits | decision  |
|-----------|---------|-----------|--------|--------|----------|-----------|
| lotsize   | 33.74   | 33.65     | 31.70  | 36.06  | 1.00     | Confirmed |
| bedrooms  | 15.01   | 15.14     | 12.88  | 17.42  | 1.00     | Confirmed |
| bathrooms | 27.38   | 27.46     | 24.69  | 30.01  | 1.00     | Confirmed |
| stories   | 19.77   | 19.67     | 18.18  | 22.16  | 1.00     | Confirmed |
| gasheat   | 3.28    | 3.48      | 0.75   | 5.30   | 0.74     | Confirmed |
| garage    | 11.21   | 11.32     | 8.20   | 13.44  | 1.00     | Confirmed |
| aircon    | 19.50   | 19.42     | 17.12  | 22.10  | 1.00     | Confirmed |

- (a) lotsize, bedrooms, bathrooms
- (b) lotsize, bathrooms, stories
- (c) gasheat, garage, aircon
- (d) lotsize, gasheat, bathrooms

2. Suppose we have a system of two structural equations as given below, where  $y_1$  and  $y_2$  are endogenous variables, and  $x_1$  and  $x_2$  are exogenous variables.

$$y_1 = \alpha_2 y_2 + \beta_1 x_1 + e_1$$

$$y_2 = \alpha_2 y_1 + \beta_2 x_2 + e_2$$

Which equation below represents a correct reduced-form equation for  $\hat{y}_1$ ?

- (a)  $\hat{y}_1 = \hat{\pi}_1 x_1 + \hat{\pi}_2 x_2 + v_1$
- (b)  $\hat{y}_1 = \hat{\pi}_1 \hat{x}_1 + \hat{\pi}_2 \hat{x}_2 + v_1$
- (c)  $\hat{y}_1 = \hat{\theta}_1 \hat{y}_2 + \hat{\pi}_1 x_1 + \hat{\pi}_2 x_2 + v_1$
- (d)  $\hat{y}_1 = \hat{\theta}_1 \hat{y}_2 + \hat{\pi}_1 \hat{x}_1 + \hat{\pi}_2 \hat{x}_2 + v_1$

3. Below we show the R code for conducting a heteroskedasticity test. The data are based on electric power generation, and the variables are total cost (**cost**, in \$), total output (**output**), and fuel price (**fuel**, in \$) . The model is given by  $cost = \beta_0 + \beta_1 output + \beta_2 fuel + e$ . Based on the code and respective output, identify which test was performed and the conclusion from the test.

```
library(car)
library(broom)
data(saving)
attach(saving)
> model = lm(cost~output + fuel)
> alpha <- 0.05
> ressq <- resid(model)^2
> # The test equation:
> modres <- lm(ressq~output + fuel + output:fuel + I(fuel^2))
> N <- nobs(modres)
> gmodres <- glance(modres)
> S <- gmodres$df
> chisqcr <- qchisq(1-alpha, S)
> print(chisqcr)
[1] 9.487729
> Rsqres <- gmodres$r.squared
> chisq <- N*Rsqres
> print(chisq)
[1] 94.87646
```

- (a) Breusch-Pagan test and conclude heteroskedasticity is present
- (b) White test and conclude heteroskedasticity is present
- (c) Breusch-Pagan test and conclude heteroskedasticity is not present
- (d) White test and conclude Heteroskedasticity is not present

4. Suppose we estimate the equation  $price = \beta_0 + \beta_1 sqft + \beta_2 bdrms + e$ , where **price** represents house prices (in, \$1000s), **sqft** is the size of the house in square feet, and **bdrms**, the number of bedrooms in the house. To further assess the model's performance, we run 5-fold CV on the model. The respective R code and CV output is given below:

```
fit= lm(price ~ sqft + bdrms,x = TRUE, y = TRUE, data=hprice1)
cv.lm(fit, k = 5)
Mean absolute error      : 48.25429
Sample standard deviation : 7.6221

Mean squared error      : 4343.904
Sample standard deviation : 1182.484

Root mean squared error  : 65.40264
Sample standard deviation : 9.110364
```

Which option below is the correct interpretation of the RMSE?

- (a) On average, the estimates of home prices are off by \$65.40k
- (b) On average, the estimates of home prices are off by 65.0%
- (c) On average, the estimates of home prices are off by \$65.40
- (d) On average, the estimates of home prices are off by \$654.0

5. We estimate the model :  $fatal = \beta_0 + \beta_1 miles + \beta_2 pop + \beta_4 income + e$ , using 336 observations from the datafile **Fatalities**. The data represent US traffic fatalities for the “lower 48” US states from 1982 to 1988. The R code below performs a Breusch-Pagan test for heteroskedasticity. Which conclusion is correct?

```
> model = lm(fatal ~ miles + pop + income, data=Fatalities)
> res2 = (model$residuals)^2
> summary(lm(res2 ~ miles + pop + income, data=Fatalities ))
```

Call:

```
lm(formula = res2 ~ miles + pop + income, data = Fatalities)
```

Residuals:

|  | Min     | 1Q     | Median | 3Q    | Max     |
|--|---------|--------|--------|-------|---------|
|  | -490944 | -52739 | -16891 | 20055 | 1420226 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | -3.502e+05 | 7.717e+04  | -4.538  | 7.96e-06 *** |
| miles       | 4.280e+01  | 6.295e+00  | 6.799   | 4.90e-11 *** |
| pop         | 1.957e-02  | 1.961e-03  | 9.983   | < 2e-16 ***  |
| income      | -1.403e+00 | 4.299e+00  | -0.326  | 0.744        |

---

Signif. codes: 0 ‘\*\*\*’ 0.001 ‘\*\*’ 0.01 ‘\*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 165000 on 332 degrees of freedom

Multiple R-squared: 0.2828, Adjusted R-squared: 0.2763

F-statistic: 43.63 on 3 and 332 DF, p-value: < 2.2e-16

```
> chi_crit = qchisq(p=.05, df=2, lower.tail=FALSE)
```

```
> print(chi_crit)
```

```
[1] 5.991465
```

- The variance model used is  $var(e) = h(\alpha_0 + \alpha_1 miles + \alpha_2 pop)$ , and based on the test, we conclude that heteroskedasticity is not present.
- The variance model used is  $var(e) = h(\alpha_0 + \alpha_1 miles + \alpha_2 pop + \alpha_3 income)$ , and based on the test, we conclude that heteroskedasticity is present.
- The variance model used is  $var(e) = h(\alpha_0 + \alpha_1 miles + \alpha_2 pop + \alpha_3 income)$ , and based on the test, we conclude that heteroskedasticity is not present.
- The variance model used is  $var(e) = h(\alpha_0 + \alpha_1 miles + \alpha_2 pop)$ , and based on the test, we conclude that heteroskedasticity is present.

6. What is the order of the  $AR(p)$  model below?

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-4} + \phi_3 y_{t-7} + \varepsilon_t$$

- (a)  $p = 3$
- (b)  $p = 12$
- (c)  $p = 7$
- (d) None of the above

7. Suppose you estimate a multiple regression model with 10 predictors,  $x_1, \dots, x_{10}$ , and one response variable  $y$ . Which plot below would you use to visually detect heteroskedasticity? Assume you do not know which predictor(s) exhibit(s) heteroskedasticity with  $y$ .

- (a) Q-Q Plot
- (b) Spread-and-Level Plot
- (c) Plot of the regression residuals vs.  $x_1$
- (d) Plot of  $x_1$  vs.  $x_{10}$

8. The inverse demand curve for truffles can be written as

$$P_i = \delta_1 Q_i + \delta_2 PS_i + \delta_3 DI_i + u^d$$

where  $PS_i$  is the price of a substitute good and  $DI_i$  is disposable income. To estimate the demand curve, we instrument for the endogenous variable  $Q_i$  with  $PF_i$  (the price of a factor of production). The output from two stage least squares estimation is:

Call:

```
ivreg(formula = p ~ q + ps + di | pf + ps + di, data = truffles)
```

Residuals:

|  | Min     | 1Q     | Median | 3Q    | Max    |
|--|---------|--------|--------|-------|--------|
|  | -39.661 | -6.781 | 2.410  | 8.320 | 20.251 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -11.428  | 13.592     | -0.841  | 0.40810      |
| q           | -2.671   | 1.175      | -2.273  | 0.03154 *    |
| ps          | 3.461    | 1.116      | 3.103   | 0.00458 **   |
| di          | 13.390   | 2.747      | 4.875   | 4.68e-05 *** |

Diagnostic tests:

|                  | df1 | df2 | statistic | p-value      |
|------------------|-----|-----|-----------|--------------|
| Weak instruments | 1   | 26  | 17.48     | 0.000291 *** |
| Wu-Hausman       | 1   | 25  | 120.03    | 4.92e-11 *** |
| Sargan           | 0   | NA  | NA        | NA           |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom

Multiple R-Squared: 0.5567, Adjusted R-squared: 0.5056

Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06

What can we conclude about the strength of our instruments?

- (a) Since  $F = 17.48 > 10$ , the instruments appear to be strong.
- (b) Since  $F = 120.03 > 10$ , the instruments appear to be strong.
- (c) Since  $F = 17.48 > 10$ , the instruments are weak.
- (d) Since  $F = 120.03 > 10$ , the instruments are weak.

9. Below we perform a Granger causality test for the S&P500 returns (`ret_gspc`) and FTSE returns (`ret_ftse`) to see which one influences the other.

```
>grangertest(ret_gspc ~ ret_ftse, order = 4)
Granger causality test

Model 1: ret_gspc ~ Lags(ret_gspc, 1:4) + Lags(ret_ftse, 1:4)
Model 2: ret_gspc ~ Lags(ret_gspc, 1:4)
      Res.Df Df      F      Pr(>F)
1      286
2      290 -4 9.2542 4.802e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> grangertest(ret_ftse ~ ret_gspc, order = 4)
Granger causality test

Model 1: ret_ftse ~ Lags(ret_ftse, 1:4) + Lags(ret_gspc, 1:4)
Model 2: ret_ftse ~ Lags(ret_ftse, 1:4)
      Res.Df Df      F Pr(>F)
1      286
2      290 -4 0.9411 0.4405
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the results, which conclusion is correct? Assume  $\alpha = 5\%$ .

- (a) S&P500 returns influence FTSE returns
- (b) FTSE returns influence S&P500 returns
- (c) S&P500 and FTSE returns influence each other
- (d) Neither S&P500 nor FTSE returns influence each other

10. Suppose you want to estimate a linear regression model for the college GPA (`colgpa`) as a function of total number of hours studied in the term (`tothrs`);

$$colgpa = \beta_1 + \beta_2 tothrs + e,$$

and discover that heteroskedasticity is present. The R code below shows the result from applying GLS to mitigate heteroskedasticity following the model  $\sigma_i^2 = \sigma^2 x_i^\gamma$ . Based on the output, what is the estimate of  $\gamma$ ?

```
>reg.mod = lm(colgpa ~ tothrs)
>ehatsq <- resid(reg.mod)^2
>sighatsq.ols <- lm(log(ehatsq)~log(tothrs))
>summary(sighatsq.ols)
```

Call:

```
lm(formula = log(ehatsq) ~ log(tothrs))
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max    |
|----------|---------|--------|--------|--------|
| -14.8151 | -0.9715 | 0.5131 | 1.4992 | 3.5371 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.65729 | 0.16449    | -3.996  | 6.56e-05 *** |
| log(tothrs) | -0.38027 | 0.04355    | -8.731  | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.186 on 4135 degrees of freedom

Multiple R-squared: 0.0181, Adjusted R-squared: 0.01787

F-statistic: 76.24 on 1 and 4135 DF, p-value: < 2.2e-16

- (a) -0.65729
- (b) 0.68368
- (c) 0.51825
- (d) -0.3803

11. Consider the demand equation

$$\ln(Q_t) = \alpha_1 + \alpha_2 \ln(P_t) + \alpha_3 \ln(Y_t) + \alpha_4 POPGRO_t + e_t,$$

where  $Q_t$  is the consumption of boneless chicken (lbs),  $P_t$  is the real price (index) of fresh chicken, and  $POPGRO_t$  is the population growth rate. Below we estimate the demand equation by 2SLS using as instruments  $\ln(EXPTS_{t-1})$  (exports of boneless chicken) and  $PF$  (real price of broiler feed). Based on the output below, what can we conclude about the overidentification restrictions?

```
> summary(iv.reg, diagnostics = TRUE)
Call:
ivreg(formula = log(q) ~ log(p) + log(y) + popgro | log(y) +
      popgro + lag(lexpts) + pf + y, data = newbroiler)
Residuals:
      Min       1Q   Median       3Q      Max
-0.078716 -0.035092 -0.004769  0.036053  0.098158
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -4.54708      1.11786  -4.068 0.000247 ***
log(p)       -0.45161      0.13767  -3.280 0.002307 **
log(y)        0.82127      0.11109   7.393 1.01e-08 ***
popgro        0.16723      0.03952   4.231 0.000153 ***

Diagnostic tests:
              df1 df2 statistic  p-value
Weak instruments    3  34     9.089 0.000147 ***
Wu-Hausman          1  35     8.486 0.006197 **
Sargan              2  NA    11.878 0.002634 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2696 on 46 degrees of freedom
Multiple R-Squared: 0.5342, Adjusted R-squared: 0.5038
Wald test: 32.27 on 3 and 46 DF, p-value: 2.204e-11
```

- (a) The additional instruments are valid
- (b) The additional instruments are not valid
- (c) The test is inconclusive
- (d) None of the above

12. Consider an AR(2) model for growth in real GDP:

$$G_t = \delta + \theta_1 G_{t-1} + \theta_2 G_{t-2} + v_t.$$

Below we estimate the respective model. Based on the output, find the value of  $G_{T+1}$ . Assume  $G_T = 0.3$ . and  $G_{T-1} = 0.6$ .

```
library(POE5Rdata)
library(dynlm)
data("okun5_aus")
attach(okun5_aus)
g_ts = ts(g,start=c(1985,2), end=c(2009,3), frequency=4)
model = dynlm(g_ts~L(g_ts,1:2))
summary(model)
```

Time series regression with "ts" data:  
Start = 1986(1), End = 2009(3)

Call:  
dynlm(formula = g\_ts ~ L(g\_ts, 1:2))

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.6428 | -0.5462 | -0.0071 | 0.5189 | 2.5147 |

Coefficients:

|               | Estimate | Std. Error | t value | Pr(> t )     |
|---------------|----------|------------|---------|--------------|
| (Intercept)   | 0.67101  | 0.16461    | 4.076   | 9.77e-05 *** |
| L(g_ts, 1:2)1 | 0.11560  | 0.10498    | 1.101   | 0.274        |
| L(g_ts, 1:2)2 | -0.08934 | 0.10552    | 0.847   | 0.399        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9309 on 92 degrees of freedom  
Multiple R-squared: 0.02344, Adjusted R-squared: -0.008759  
F-statistic: 0.7279 on 4 and 92 DF, p-value: 0.5379

- (a) 0.0883
- (b) -0.0189
- (c) 0.7593
- (d) 0.6521

13. Below we estimate the First Stage Least Squares for an endogenous variable  $x$ , using  $z_1$  and  $z_2$  as instruments. Based on the output below, which statement about the instruments is correct?

```
> x_hat <- lm(x ~ z1+z2, data = df)
> summary(x_hat)
```

Call:

```
lm(formula = x ~ z1 + z2, data = df)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -0.26236 | -0.06777 | -0.00258 | 0.06921 | 0.30021 |

Coefficients:

|             | Estimate  | Std. Error | t value | Pr(> t )     |
|-------------|-----------|------------|---------|--------------|
| (Intercept) | 4.511e+00 | 1.960e-02  | 230.120 | < 2e-16 ***  |
| z1          | 3.377e-02 | 3.963e-03  | 8.522   | 2.73e-13 *** |
| z2          | 2.214e-10 | 9.648e-11  | 2.295   | 0.064        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1078 on 93 degrees of freedom

Multiple R-squared: 0.5166, Adjusted R-squared: 0.5062

F-statistic: 49.7 on 2 and 93 DF, p-value: 2.088e-15

- (a)  $z_2$  is a better instrument than  $z_1$  since it is not correlated with  $x$
- (b) Both instruments are valid
- (c) Neither instrument is valid
- (d)  $z_1$  is a better instrument than  $z_2$  since it is correlated with  $x$

14. Below we estimate a wage equation using OLS and the White robust standard errors.

$$\begin{array}{rccccc}
 \widehat{WAGE} & = & -17.77 & + 2.5 EDUC & + 0.23 EXPER & + 3.23 METRO & - 4.20 FEMALE \\
 (se) & = & (2.360) & (0.140) & (0.310) & (1.050) & (0.810) \\
 (white) & = & (2.500) & (0.160) & (0.029) & (0.840) & (0.800)
 \end{array}$$

For which coefficients (ignoring the  $y$ -intercept) have interval estimates gotten narrower?

- (a) *EXPER*, *METRO*, and *FEMALE*
  - (b) *EDUC*
  - (c) *EDUC*, *EXPER*, *METRO*, and *FEMALE*
  - (d) All of them
  - (e) None of the above
15. In the R code below, we estimate an ARDL(p,q) model. Based on the R code, identify which model is being estimated?

```
model = dynlm(y~L(y,1:2)+L(x,0:1)+L(x,3))
```

- (a)  $y = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 x_{t-1} + \beta_4 x_{t-3} + e_t$
- (b)  $y = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 x_t + \beta_4 x_{t-3} + e_t$
- (c)  $y = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 x_t + \beta_4 x_{t-1} + \beta_5 x_{t-3} + e_t$
- (d)  $y = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 x_{t-3} + e_t$

16. The dataset `br5` from the `P0E5Rdata` library contains 1080 observations of houses sold in Baton Rouge, Louisiana during mid-2005. We will be concerned with the selling price (*PRICE*, measured in dollars), number of bedrooms (*BEDROOMS*), and the age of the house in years (*AGE*). Below we estimate the model

$$PRICE = \beta_1 + \beta_2 AGE + \beta_3 AGE^2 + \beta_4 SQFT + \beta_5 BEDROOMS + e,$$

using GLS for the ‘known form of the variance’ case ( $weight = 1/\sqrt{age}$ ). What is the estimated 95% confidence interval for  $\beta_4$ ? Assume  $z_c = 1.96$ .

```
library(P0E5Rdata)
library(car)
library(broom)
library(lmtest)
data(br5)
w = 1/br5$age
reg.mod = lm(price ~ age + I(age^2) + sqft + bedrooms, data=br5)
cov1 <- hccm(reg.mod, type="hc1")
reg.mod.wls = lm(price ~ age + I(age^2) + sqft + bedrooms, weights=w, data=br5)
vcv <- coeftest(reg.mod.wls, vcov.=cov1)
vcv
t test of coefficients:
```

|             | Estimate    | Std. Error | t value | Pr(> t )      |
|-------------|-------------|------------|---------|---------------|
| (Intercept) | -18.3322398 | 16.1079104 | -1.1381 | 0.2554        |
| age         | -0.1616030  | 0.3366829  | -0.4800 | 0.6314        |
| I(age^2)    | -0.0081348  | 0.0057545  | -1.4136 | 0.1578        |
| sqft        | 15.4364229  | 1.3493889  | 11.4396 | < 2.2e-16 *** |
| bedrooms    | -54.6942652 | 9.4499623  | -5.7878 | 9.862e-09 *** |

---

Signif. codes: 0 ‘\*\*\*’ 0.001 ‘\*\*’ 0.01 ‘\*’ 0.05 ‘.’ 0.1 ‘ ’ 1

- (a) [13.22, 17.66]
- (b) [-73.21, -36.17]
- (c) [12.79, 18.08]
- (d) [-70.24, -39.15]
- (e) None of the above

17. An equation in the simultaneous equations model satisfies the order condition for identification if:
- (a) the number of excluded endogenous variables from the equation is at most as large as the number of right-hand side exogenous variables.
  - (b) the number of excluded exogenous variables from the equation is at least as large as the number of right-hand side endogenous variables.
  - (c) the number of excluded endogenous variables from the equation is at least as large as the number of right-hand side exogenous variables.
  - (d) the number of excluded exogenous variables from the equation is at most as large as the number of right-hand side endogenous variables.

18. The R output below is based on a VAR model using the variables  $y_1$  and  $y_2$ . Based on the output, which statement is correct?

Estimation results for equation y2:

=====

y2 = y1.l1 + y2.l1 + y1.l2 + y2.l2 + y1.l3 + y2.l3 + const

|       | Estimate | Std. Error | t value | Pr(> t )     |
|-------|----------|------------|---------|--------------|
| y1.l1 | 0.01176  | 0.08886    | 0.132   | 0.894778     |
| y2.l1 | 0.65636  | 0.05543    | 11.841  | < 2e-16 ***  |
| y1.l2 | -0.11930 | 0.08399    | -1.420  | 0.156410     |
| y2.l2 | 0.23870  | 0.06476    | 3.686   | 0.000266 *** |
| y1.l3 | -0.01911 | 0.08164    | -0.234  | 0.815015     |
| y2.l3 | 0.14353  | 0.05930    | 2.420   | 0.016048 *   |
| const | 0.13086  | 0.03577    | 3.658   | 0.000295 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1184 on 332 degrees of freedom

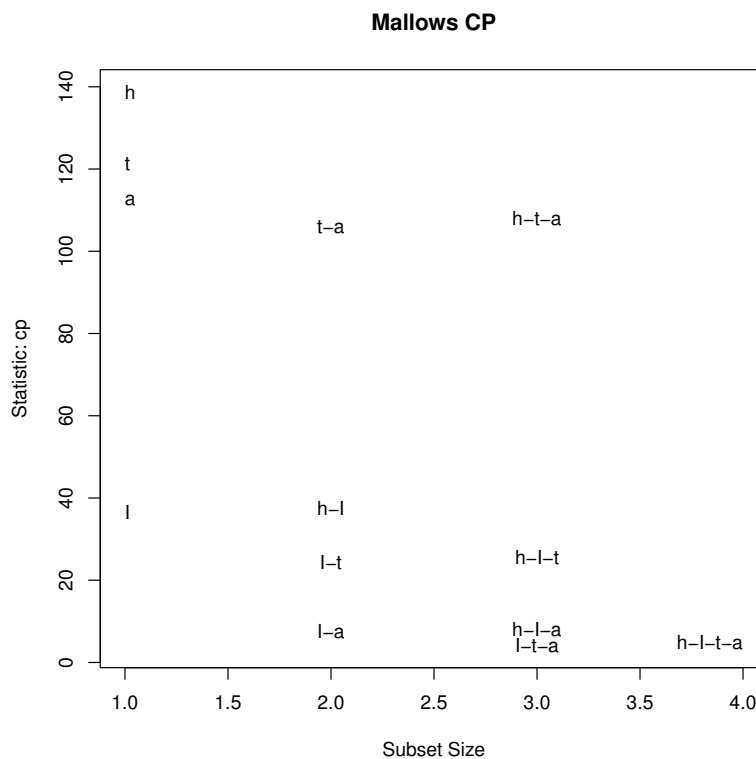
Multiple R-Squared: 0.8958, Adjusted R-squared: 0.894

F-statistic: 475.9 on 6 and 332 DF, p-value: < 2.2e-16

- (a) It appears that only lags of  $y_2$  influence  $y_2$
- (b) The VAR results are inconclusive
- (c) It appears that  $y_1$  influences  $y_2$
- (d) It appears that  $y_1$  and  $y_2$  influence each other

19. For this problem, the response variable is monthly earnings (in \$, **wage**), and the predictors are, average weekly hours worked (**hours**), age in years (**age**), IQ score (**IQ**), and years with current employer (**tenure**). Based on the Mallows Cp plot below, which model is the best one if you had to choose two predictors only?

|        | Abbreviation |
|--------|--------------|
| hours  | h            |
| IQ     | I            |
| tenure | t            |
| age    | a            |



- (a) I-a
- (b) t-a
- (c) h-I
- (d) I-t

20. Suppose we estimate a model for cumulative GPA (*cumgpa*) as a function of SAT score (*sat*), percentile in high school (*hsperc*), and gender (*female*: =1 if female). Test for heteroskedasticity using a Breusch-Pagan test and the variables *sat*, *hsperc*, and *tothrs*. Is there evidence of heteroskedasticity at a 5% level of significance? Assume the variance of the residuals follows the model  $var(e) = \alpha_1 + \alpha_2 sat + \alpha_3 hsperc + \alpha_4 tothrs$ . Below is the respective residual regression output.

Call:

```
lm(formula = res2 ~ sat + hsperc + tothrs)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.5458 | -0.6514 | -0.1340 | 0.3853 | 6.9477 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 2.1607914  | 0.2850160  | 7.581   | 1.04e-13 *** |
| sat         | -0.0003189 | 0.0002592  | -1.230  | 0.21899      |
| hsperc      | -0.0061824 | 0.0019552  | -3.162  | 0.00163 **   |
| tothrs      | -0.0194507 | 0.0011615  | -16.746 | < 2e-16 ***  |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.081 on 728 degrees of freedom

Multiple R-squared: 0.2815, Adjusted R-squared: 0.2785

F-statistic: 95.06 on 3 and 728 DF, p-value: < 2.2e-16

- $H_0 : \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0$ . Since  $\chi^2 = 206.06 > \chi_{crit}^2 = 5.99$ , we fail reject  $H_0$  at the 5% level of significance, and conclude that heteroskedasticity is not present.
- $H_0 : \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0$ . Since  $\chi^2 = 5.99 < \chi_{crit}^2 = 206.06$ , we fail reject  $H_0$  at the 5% level of significance, and conclude that heteroskedasticity is not present.
- $H_0 : \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0$ . Since  $\chi^2 = 5.99 < \chi_{crit}^2 = 206.06$ , we reject  $H_0$  at the 5% level of significance, and conclude that heteroskedasticity is present.
- $H_0 : \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0$ . Since  $\chi^2 = 206.06 > \chi_{crit}^2 = 5.99$ , we reject  $H_0$  at the 5% level of significance, and conclude that heteroskedasticity is present.
- None of the above